

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY
INFORMATION SYSTEMS AND TECHNOLOGY DEPARTMENT
TEST 2, INTRODUCTION TO OPERATING SYSTEM - ET 8210
UQF 8 SECOND YEAR BACHELOR, ETE

14/20

TIME: 1 HR,

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ANSWER ALL QUESTIONS

OBSERVE AND ABIDE TO ALL UNIVERSITY EXAMINATIONS REGULATIONS

In questions (1-10) choose the most correct answer and write its letter [01 mark]

1. What is Scheduling?
a) allowing a job to use the processor b) making proper use of processor
☒ c) all of the mentioned d) none of the mentioned
2. A set of processes is deadlock if _____
☒ a) each process is blocked and will remain so forever
b) each process is terminated
c) all processes are trying to kill each other d) none of the mentioned
3. The strategy of making processes that are logically runnable to be temporarily suspended is called _____
☒ a) Non preemptive scheduling b) Preemptive scheduling B.
c) Shortest job first d) First come First served
4. Which is the most optimal scheduling algorithm?
a) FCFS - First come First served b) SJF - Shortest Job First
☒ c) RR - Round Robin d) None of the mentioned
5. The processes that are residing in main memory and are ready and waiting to execute are kept on a list called _____
a) job queue ☒ b) ready queue c) execution queue d) process queue
6. The interval from the time of submission of a process to the time of completion is termed as _____
☒ a) waiting time b) turnaround time c) response time d) throughput B.

7. In operating system, each process has its own _____
 a) address space and global variables b) open files
 c) pending alarms, signals and signal handlers ☒ d) all of the mentioned
8. The switching of the CPU from one process or thread to another is called _____
 a) process switch b) task switch c) context switch ☒ d) all of the mentioned
9. What is Dispatch latency?
 a) the speed of dispatching a process from running to the ready state
 b) the time of dispatching a process from running to ready state and keeping the CPU idle
☒ c) the time to stop one process and start running another one
 d) none of the mentioned
10. Choose one of the disadvantages of the priority scheduling algorithm?
 a) it schedules in a very complex manner b) its scheduling takes up a lot of time
 c) it can lead to some low priority process waiting indefinitely for the CPU ☒ d) none of the mentioned

11. You are provided with the processes below in table 1

[8 marks]

PROCESS	CPU burst time (ms)
P1	2
P2	22
P3	14
P4	31
P5	7

Table 1.

- Using FCFS - FIFO, draw Gantt chart hence calculate average waiting time, average turnaround time and average throughput
- Using Shortest Job First (SJF) Scheduling, draw Gantt chart hence calculate average waiting time, average turnaround time and average throughput

12. How is noncontiguous allocation done?

[2 marks]

12. Noncontiguous allocation is done into two methods paging and segmentation and divides its blocks into either pages or segments which are subsequently allocated to different memory locations in accordance with the quantity of free memory.

BONIFACE SANKWA

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By using FCFS - FIFO

Grantt. chart

P_1	P_2	P_3	P_4	P_5	
0	2	24	38	69	76

Average waiting time

waiting time of

$$P_1 = 0 \text{ ms}$$

$$P_2 = (2 - 0) \text{ ms} = 2 \text{ ms}$$

$$P_3 = (24 - 0) \text{ ms} = 24 \text{ ms}$$

$$P_4 = (38 - 0) \text{ ms} = 38 \text{ ms}$$

$$P_5 = (69 - 0) \text{ ms} = 69 \text{ ms}$$

$$\text{Average waiting time} = \frac{(0 + 2 + 24 + 38 + 69) \text{ ms}}{5} = 26.6 \text{ ms}$$

Average waiting time is 26.6 ms. ✓

Average turnaround time

turnaround time of

$$P_1 = 0 + 2 = 2 \text{ ms}$$

$$P_2 = 2 + 22 = 24 \text{ ms}$$

$$P_3 = 24 + 14 = 38 \text{ ms}$$

$$P_4 = 38 + 31 = 69 \text{ ms}$$

$$P_5 = 69 + 7 = 76 \text{ ms}$$

Average waiting time

$$\text{Average turnaround time} = \frac{(2 + 24 + 38 + 69 + 76) \text{ ms}}{5} = 41.8 \text{ ms}$$

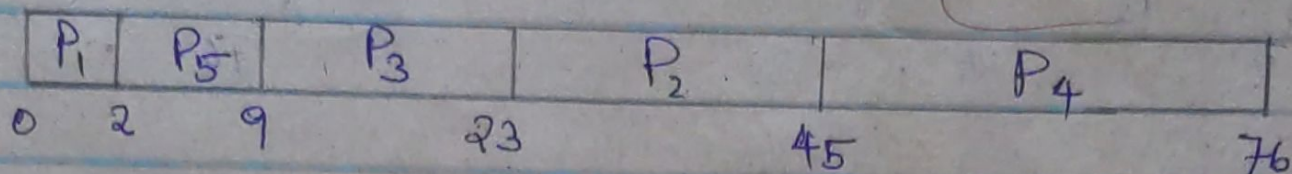
average turnaround time is 41.8 ms. ✓

$$\text{Average throughput} = \frac{(2 + 22 + 14 + 31 + 7) \text{ ms}}{5} = 15.2 \text{ ms}$$

Average throughput is 15.2 ms. ✓

11 (ii) By using shortest Job First (SJF)

Gantt chart



• Average waiting time
waiting time of

$$P_1 = 0 \text{ ms}$$

$$P_2 = (23 - 0) \text{ ms} = 23 \text{ ms}$$

$$P_3 = (9 - 0) \text{ ms} = 9 \text{ ms}$$

$$P_4 = (45 - 0) \text{ ms} = 45 \text{ ms}$$

$$P_5 = (2 - 0) \text{ ms} = 2 \text{ ms}$$

$$\text{Average waiting time} = \frac{(0 + 23 + 9 + 45 + 2) \text{ ms}}{5} = 15.8 \text{ ms}$$

$$\text{Average waiting time is } \underline{15.8 \text{ ms.}}$$

• Average turnaround time
Turnaround time of

$$P_1 = 0 + 2 = 2 \text{ ms}$$

$$P_2 = 23 + 22 = 45 \text{ ms}$$

$$P_3 = 9 + 14 = 23 \text{ ms}$$

$$P_4 = 45 + 31 = 76 \text{ ms}$$

$$P_5 = 2 + 7 = 9 \text{ ms}$$

$$\text{Average turnaround time} = \frac{(2 + 45 + 23 + 76 + 9) \text{ ms}}{5} = 31 \text{ ms}$$

$$\text{Average turnaround time is } \underline{31 \text{ ms.}}$$

$$\text{Average throughput} = \frac{(2 + 22 + 14 + 31 + 7) \text{ ms}}{5}$$

$$\text{Average throughput is } \underline{15.2 \text{ ms}}$$